LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Study of pressure distribution over an aerofoil at various angles of attack, determination of center of pressure and aerodynamic center.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Study of pressure distribution over an aerofoil at various angles of attack, determination of center of pressure and aerodynamic center. various forces e.g. lift, drag and moments
5. Specific Objectives	6.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Analysis of pressure distribution graphs for different angles of attack; identification of CP and AC locations; safety: Handle graph sheets carefully; supervision: Instructor monitors graph interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary forces acting on an aerofoil?
(b) How does the angle of attack affect lift generation?
(c) Define the term 'aerofoil section'.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Pressure distribution at low angles of attack • Pressure distribution at high angles of attack • Variation of center of pressure location • Definition of aerodynamic center • Relationship between lift and moment coefficients	L & Ds	PPT, IB	15 min
(b)	• Calculation of center of pressure shift • Constant moment point concept • Impact of angle of attack on stability • Graphical determination of CP • Comparison of CP and AC positions	L & Ds	PPT, PDF	20 min
(c)	• Practical analysis of force vectors • Identification of lift and drag components • Moment arm determination techniques • Effect of camber on pressure distribution • Summary of performance conditions	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why the aerodynamic center remains constant while the center of pressure moves.
(b) Describe how pressure distribution changes as the angle of attack increases towards stall.
(c) How does the location of the center of pressure affect the pitching moment of the aerofoil?

12. Home Assignment and Follow Up	Prepare a brief note comparing the NACA series aerofoil characteristics discussed in the next lesson with the general pressure distribution concepts covered today, focusing on how shape influences the center of pressure.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	NACA series, Vortex theory, Magnus effect, Kutta Jakubowski law, Span wise flow, down Wash wing tip and trailing edge vortices. Methods for preventing span-wise flow.	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min